

RMO(INDIA) – 1987(1)

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Prove that the expression $2x+3y$ and $9x+5y$ are divisible by 17 for the same set of integral values of x and y .
 2. Prove that there exist $2^n - 2$ numbers that have n digits made up only of the numbers 1 and 2 and contain each digit at least once.
 3. The points A_0, A_1, A_2, A_3, A_4 divide a unit circle into five equal parts. Prove that the chords A_0A_1, A_0A_2 satisfy $(A_0A_1 \times A_0A_2)^2 = 5$.
 4. If two triangles have a common angle then prove that the sum of the sines of the angles will be larger in that triangle where the difference of the remaining two angles is smaller.
 5. Prove that for any natural number n the expression $A = 2903^n - 803^n - 464^n + 261^n$ is divisible by 1897.
 6. Solve the system of equations $x + y + z = a$, $x^2 + y^2 + z^2 = b^2$, $xy = z^2$ where a and b are constants. Give the conditions on a and b so that x, y and z are distinct positive numbers.
 7. Let a, b, c be real numbers such that $(bc - a^2)^{-1} + (ca - b^2)^{-1} + (ab - c^2)^{-1} = 0$. Prove that $a(bc - a^2)^{-2} + b(ca - b^2)^{-2} + c(ab - c^2)^{-2} = 0$.
 8. Let A_1A_2, B_1B_2, C_1C_2 be three equal segments on the sides of an equilateral triangle. Prove that in the triangle formed by the lines B_2C_1, C_2A_1 and A_2B_1 the segments B_2C_1, C_2A_1 and A_2B_1 are proportional to the sides in which they are contained.
 9. Determine all positive integers n for which $2^n + 1$ is divisible by 3.

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